2025 年度 科学技術計算 第 3 回(通常) レポート

学籍番号　36714121

堀航輔

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1.はじめに

このレポートでは、行列と連立一次方程式の課題について述べる。プログラムの説明や動作の他、問題に対する回答などが含まれる。なお、この課題において利用した Python のバージョンは全て 3.11.9 である。

2.演習課題

2.1 課題 1

ガウス消去法で一次連立方程式を解くプログラムを 2 種類作成する。ピボッティングを「行わない」、または「行った」ガウス消去法による解法である。

プログラム名は” Gauss_nopivot.py”及び”Gauss.py”である。どちらも、ベクトルの定義や拡大行列を作った後、ピポット要素や対角要素などに注意しつつ、前進消去と後進代入を行ったのち、結果を出力する。もしも途中で一意解が得られない問題だと判断された場合は sys により、その旨を出力してプログラムが終了する。また、後者のプログラムでは、前進消去中にピボッティングを行い、システムの停止を回避する。

以下はテストに対するプログラムの動作の様子と結果である。

```
# Gauss_nopivot.py(上から(1),(2),(3)のデータをテスト)
初期値(1)
[[ 8.2     3.8    28.14]
[ 4.1     1.4    11.77]]

k =    0 , pivot =    8.2
[[ 8.2     3.8    28.14]
[ 0.     -0.5    -2.3 ]]

解  x =    [1.3 4.6]

初期値(2)
[[ 4.    1. -1.    1.]
[ 8.    2.    3. 12.]
[ 1.    3.    4.    6.]]

k =    0 , pivot =    4.0
```

```
[[ 4.       1.     -1.       1.    ]
[ 0.       0.       5.     10.    ]
[ 0.       2.75    4.25    5.75]]

k =    1 , pivot =    0.0
```
一意解は存在しない

初期値(3)
```
[[    1.2    -0.9    -4. ]
[ -8.4      6.3 -28. ]]

k =    0 , pivot =    1.2
[[ 1.2000000e+00 -9.0000000e-01 -4.0000000e+00]
[ 0.0000000e+00 -8.8817842e-16 -5.6000000e+01]]
```

一意解は存在しない

Gauss.py(上から(1),(2),(3)のデータをテスト)

初期値(1)
```
[[ 8.2      3.8    28.14]
[ 4.1      1.4    11.77]]

k =    0 , pivot =    8.2
[[ 8.2      3.8    28.14]
[ 0.      -0.5    -2.3 ]]

解  x =    [1.3 4.6]
```

初期値(2)
```
[[ 4.    1. -1.    1.]
[ 8.    2.    3. 12.]
[ 1.    3.    4.    6.]]
```

k = 0 , pivot = 4.0

[[8. 2. 3. 12.]

[0. 0. -2.5 -5.]

[0. 2.75 3.625 4.5]]

k = 1 , pivot = 0.0

[[8. 2. 3. 12.]

[0. 2.75 3.625 4.5]

[0. 0. -2.5 -5.]]

解 x = [1. -1. 2.]

初期値(3)

[[1.2 -0.9 -4.]

[-8.4 6.3 -28.]]

k = 0 , pivot = 1.2

[[-8.40000000e+00 6.30000000e+00 -2.80000000e+01]

[0.00000000e+00 -1.11022302e-16 -8.00000000e+00]]

一意解は存在しない(ピポッティング失敗)

2.2 課題 2

(1)ヤコビ法、およびガウスザイデル法を実装する。

プログラムについて説明する。プログラム名は”Jacobi.py”及び”Zidel.py”である。ヤコビ法については、あらかじめ行列を対角成分が 1 になるよう変形しておき、それをヤコビ法の公式に則って計算していく。この時差分を逐次計算し、差分が一定値を下回った時にループを抜けて結果を出力する。ガウスザイデル法は、ヤコビ法から、行ごとに値を行うようにして更新後の値を直接代入して次の行の計算を行うことで実装した。

(2) ヤコビ法とガウスザイデル反復法の収束の早さの違いの考察。反復回数を比較する。

以下に実行結果を示す。

Jacobi.py(上から、講義資料、(1)、(2))

A = [[5. -4.]

[-1. 2.]]

b = [4. 1.]

x0 = [0. 0.]

k=00: x=[0. 0.], d=-

k=01: x=[0.8 0.5], d=0.800000

k=02: x=[1.2 0.9], d=0.400000

k=03: x=[1.52 1.1], d=0.320000

k=04: x=[1.68 1.26], d=0.160000

k=05: x=[1.808 1.34], d=0.128000

k=06: x=[1.872 1.404], d=0.064000

k=07: x=[1.9232 1.436], d=0.051200

k=08: x=[1.9488 1.4616], d=0.025600

k=09: x=[1.96928 1.4744], d=0.020480

k=10: x=[1.97952 1.48464], d=0.010240

k=11: x=[1.987712 1.48976], d=0.008192

k=12: x=[1.991808 1.493856], d=0.004096

k=13: x=[1.9950848 1.495904], d=0.003277

k=14: x=[1.9967232 1.4975424], d=0.001638

k=15: x=[1.99803392 1.4983616], d=0.001311

k=16: x=[1.99868928 1.49901696], d=0.000655

k=17: x=[1.99921357 1.49934464], d=0.000524

k=18: x=[1.99947571 1.49960678], d=0.000262

k=19: x=[1.99968543 1.49973786], d=0.000210

k=20: x=[1.99979028 1.49984271], d=0.000105

k=21: x=[1.99987417 1.49989514], d=0.000084

x = [1.99987417 1.49989514]

d = 0.000084

count = 21

A = [[4.1 1.4]

[8.2 3.8]]

b = [11.77 28.14]

x0 = [0. 0.]

k=00: x=[0. 0.], d=-

k=01: x=[2.87073171 7.40526316], d=7.405263

k=02: x=[0.34210526 1.21052632], d=6.194737

k=03: x=[2.45738126 6.66703601], d=5.456510

k=04: x=[0.59418283 2.10249307], d=4.564543

k=05: x=[2.15280724 6.12307917], d=4.020586

k=06: x=[0.77992419 2.75973174], d=3.363347

k=07: x=[1.92838428 5.72226886], d=2.962537

k=08: x=[0.91678624 3.24401286], d=2.478256

k=09: x=[1.76302 5.42693495], d=2.182922

k=10: x=[1.01763197 3.60085158], d=1.826083

k=11: x=[1.64117263 5.20932049], d=1.608469

k=12: x=[1.09193935 3.86378538], d=1.345535

k=13: x=[1.55139036 5.04897299], d=1.185188

k=14: x=[1.14669215 4.05752607], d=0.991447

k=15: x=[1.485235 4.9308222], d=0.873296

k=16: x=[1.18703632 4.20028236], d=0.730540

k=17: x=[1.43648895 4.84376373], d=0.643481

k=18: x=[1.2167636 4.30547122], d=0.538293

k=19: x=[1.4005708 4.77961538], d=0.474144

k=20: x=[1.23866792 4.38297879], d=0.396637

k=21: x=[1.3741048 4.73234817], d=0.349369

k=22: x=[1.25480794 4.44008964], d=0.292259

k=23: x=[1.35460354 4.69751971], d=0.257430

k=24: x=[1.26670059 4.48217131], d=0.215348

k=25: x=[1.34023419 4.67185663], d=0.189685

k=26: x=[1.27546359 4.51317886], d=0.158678

k=27: x=[1.32964624 4.65294699], d=0.139768

k=28: x=[1.28192054 4.53602653], d=0.116920

k=29: x=[1.3218446 4.63901357], d=0.102987

k=30: x=[1.28667829 4.55286165], d=0.086152

k=31: x=[1.31609602 4.62874684], d=0.075885

k=32: x=[1.29018401 4.56526648], d=0.063480

k=33: x=[1.31186023 4.62118188], d=0.055915

k=34: x=[1.29276716 4.57440688], d=0.046775

k=35: x=[1.30873911 4.6156077], d=0.041201

k=36: x=[1.29467054 4.58114191], d=0.034466

k=37: x=[1.30643935 4.61150041], d=0.030359

k=38: x=[1.29607303 4.58610457], d=0.025396

k=39: x=[1.30474478 4.60847399], d=0.022369

k=40: x=[1.29710644 4.58976126], d=0.018713

k=41: x=[1.30349616 4.60624399], d=0.016483

k=42: x=[1.29786791 4.59245567], d=0.013788

k=43: x=[1.30257611 4.60460084], d=0.012145

k=44: x=[1.29842898 4.59444102], d=0.010160

k=45: x=[1.30189819 4.60339009], d=0.008949

k=46: x=[1.29884241 4.59590391], d=0.007486

k=47: x=[1.30139867 4.60249796], d=0.006594

k=48: x=[1.29914704 4.59698183], d=0.005516

k=49: x=[1.3010306 4.6018406], d=0.004859

k=50: x=[1.2993715 4.59777608], d=0.004065

k=51: x=[1.30075939 4.60135623], d=0.003580

k=52: x=[1.2995369 4.59836132], d=0.002995

k=53: x=[1.30055955 4.60099933], d=0.002638

k=54: x=[1.29965877 4.59879255], d=0.002207

k=55: x=[1.3004123 4.60073635], d=0.001944

k=56: x=[1.29974856 4.5991103], d=0.001626

k=57: x=[1.3003038 4.60054257], d=0.001432

k=58: x=[1.29981473 4.59934443], d=0.001198

k=59: x=[1.30022385 4.60039979], d=0.001055

k=60: x=[1.29986349 4.59951695], d=0.000883

k=61: x=[1.30016494 4.60029458], d=0.000778

k=62: x=[1.29989941 4.59964407], d=0.000651

k=63: x=[1.30012154 4.60021706], d=0.000573

k=64: x=[1.29992588 4.59973774], d=0.000479

k=65: x=[1.30008955 4.60015994], d=0.000422

k=66: x=[1.29994539 4.59980675], d=0.000353

k=67: x=[1.30006599 4.60011785], d=0.000311

k=68: x=[1.29995976 4.59985761], d=0.000260

k=69: x=[1.30004862 4.60008684], d=0.000229

k=70: x=[1.29997035 4.59989508], d=0.000192

k=71: x=[1.30003583 4.60006399], d=0.000169

k=72: x=[1.29997815 4.59992269], d=0.000141

k=73: x=[1.3000264 4.60004715], d=0.000124

k=74: x=[1.2999839 4.59994303], d=0.000104

k=75: x=[1.30001945 4.60003474], d=0.000092

x = [1.30001945 4.60003474]

d = 0.000092

count = 75

A = [[8. 2. 3.]

 [4. 1. -1.]

 [1. 3. 4.]]

b = [12. 1. 6.]

x0 = [0. 0. 0.]

k=00: x=[0. 0. 0.], d=-

k=01: x=[1.5 1. 1.5], d=1.500000

k=02: x=[0.6875 -3.5 0.375], d=4.500000

k=03: x=[2.234375 -1.375 3.953125], d=3.578125

k=04: x=[0.36132812 -3.984375 1.97265625], d=2.609375

k=05: x=[1.75634766 1.52734375 4.39794922], d=5.511719

k=06: x=[-0.53106689 -1.62744141 -0.08459473], d=4.482544

k=07: x=[1.93858337 3.03967285 2.85334778], d=4.667114

k=08: x=[-0.32992363 -3.90098572 -1.26440048], d=6.940659

k=09: x=[2.94939661 1.05529404 4.5082202], d=5.772621

k=10: x=[-0.45440608 -6.28936625 -0.02881968], d=7.344660

k=11: x=[3.08314894 2.78880465 6.3306262], d=9.078171

k=12: x=[-1.57118599 -5.00196956 -1.36239072], d=7.790774

k=13: x=[3.26138891 5.92235323 5.64427367], d=10.924323

k=14: x=[-2.09719093 -6.40128198 -3.75711215], d=12.323635

k=15: x=[4.50923755 5.63165158 6.82525922], d=12.032934

k=16: x=[-2.4673851 -10.21169099 -3.85104808], d=15.843343

k=17: x=[5.49706578 7.01849233 9.77561452], d=17.230183

k=18: x=[-3.92047853 -11.21264858 -5.13813569], d=18.231141

k=19: x=[6.22996303 11.54377842 10.88960607], d=22.756427

k=20: x=[-5.46954688 -13.03024605 -8.71532457], d=24.574024

k=21: x=[8.02580823 14.16286295 12.64007126], d=27.193109

k=22: x=[-6.78074246 -18.46316165 -11.12859927], d=32.626025

k=23: x=[10.28901514 16.99437057 17.04255685], d=35.457532

k=24: x=[-9.13955146 -23.1135037 -13.81803171], d=40.107874

k=25: x=[12.46013782 23.74017414 21.12001564], d=46.853678

k=26: x=[-12.3550494 -27.72053563 -19.42016506], d=51.460710

k=27: x=[15.7126958 31.00003255 25.37916407], d=58.720568

k=28: x=[-15.76719466 -36.47161914 -25.67819836], d=67.471652

k=29: x=[20.24722917 38.3905803 32.79551302], d=74.862199

k=30: x=[-20.39596246 -47.19340366 -32.35474252], d=85.583984

k=31: x=[25.43137936 50.22910732 41.99404336], d=97.422511

k=32: x=[-26.80504309 -58.73147407 -42.52967533], d=108.960581

k=33: x=[32.13149677 65.69049703 52.24986633], d=124.421971

k=34: x=[-34.51632413 -75.27612074 -55.80074696], d=140.966618

k=35: x=[41.2443103 83.26454956 66.58617159], d=158.540670

k=36: x=[-44.28595173 -97.3910696 -71.25948974], d=180.655619

k=37: x=[52.57007605 106.88431719 85.61479013], d=204.275387

k=38: x=[-57.3266256 -123.66551408 -91.80575691], d=230.549831

k=39: x=[66.84353736 138.50074548 108.58079196], d=262.166260

k=40: x=[-73.84298335 -157.79335748 -119.08644345], d=296.294103

k=41: x=[85.60575567 177.28548997 138.30576395], d=335.078847

k=42: x=[-94.68603397 -203.11725871 -152.86555639], d=380.402749

k=43: x=[109.60389832 226.8785795 177.50945253], d=429.995838

k=44: x=[-121.78568957 -259.90614077 -196.05990921], d=486.784720

k=45: x=[139.99900115 292.08284908 226.87602797], d=551.988990

k=46: x=[-156.59922276 -332.11997661 -252.5618871], d=624.202826

k=47: x=[179.24070181 374.83500394 289.73978815], d=706.954981

k=48: x=[-200.86117154 -426.22301911 -324.43642841], d=801.058023

k=49: x=[229.71941543 480.00825775 371.38255722], d=906.231277

k=50: x=[-257.7705234 -546.49510451 -415.93604717], d=1026.503362

k=51: x=[294.09979382 616.14604641 475.81395923], d=1162.641151

k=52: x=[-330.96674631 -699.58521604 -534.13448326], d=1315.731262

k=53: x=[376.69673523 790.73250199 608.93059861], d=1490.317718

k=54: x=[-424.53209998 -896.85634232 -685.7235603], d=1687.588844

k=55: x=[482.86042069 1013.4048396 780.27528173], d=1910.261182

k=56: x=[-544.45444055 -1150.16640104 -879.26873487], d=2163.571241

k=57: x=[618.76737584 1299.54902733 1000.23841092], d=2449.715428

k=58: x=[-698.47666093 -1473.83109243 -1127.85361446], d=2773.380120

k=59: x=[792.90287853 1667.05302925 1281.49248456], d=3140.884122

k=60: x=[-895.82293902 -1889.11902956 -1447.01549157], d=3556.172059

k=61: x=[1016.41056673 2137.27626451 1642.29500692], d=4026.395294

k=62: x=[-1148.67969372 -2422.34725999 -1855.55984007], d=4559.623524

k=63: x=[1302.92175502 2740.15893483 2105.43036842], d=5162.506195

k=64: x=[-1473.07612187 -3105.25665166 -2379.34963988], d=5845.415586

k=65: x=[1670.07027787 3513.95484758 2698.71151922], d=6619.211499

k=66: x=[-1889.0055316 -3980.56959227 -3051.48370516], d=7494.524440

k=67: x=[2140.9487875 4505.53842125 3459.1785771], d=8486.108014

k=68: x=[-2422.07657173 -5103.6165729 -3912.89101281], d=9609.154994

k=69: x=[2744.73827303 5776.41527409 4434.73157261], d=10880.031847

k=70: x=[-3105.62815825 -6543.22151951 -5016.99602383], d=12319.636794

k=71: x=[3518.67888881 7406.51660918 5685.3231792], d=13949.738129

k=72: x=[-3982.12534449 -8388.39237606 -6433.05717909], d=15794.908985

k=73: x=[4510.99453617 9496.44419889 7288.32561816], d=17884.836575

k=74: x=[-5105.73315653 -10754.65252652 -8248.58178321], d=20251.096725

k=75: x=[5783.38130033 12175.35084293 9343.92268402], d=22930.003369

k=76: x=[-6546.30871724 -13788.60251731 -10575.85845728], d=25963.953360

k=77: x=[7414.59755081 15610.37641168 11979.52906729], d=29398.978929

k=78: x=[-8393.41750316 -17677.86113593 -13559.93169646], d=33288.237548

k=79: x=[9505.93967016 20014.73831616 15358.25022774], d=37692.599452

k=80: x=[-10761.52841444 -22664.50845289 -17386.03865466], d=42679.246769

k=81: x=[12187.39160872 25661.07500311 19690.26344328], d=48325.583456

k=82: x=[-13797.61754201 -29058.3029916 -22291.15415451], d=54719.377995

k=83: x=[15625.25855584 32900.31601352 25244.6316292], d=61958.619005

k=84: x=[-17690.31586433 -37255.40259417 -28580.0516491], d=70155.718608

k=85: x=[20032.87001695 42182.21180822 32365.63091171], d=79437.614402

k=86: x=[-22681.16454395 -47764.8491561 -36643.3763604], d=89947.060964

k=87: x=[25683.97842418 54082.28181538 41495.42800306], d=101847.130971

k=88: x=[-29079.85595499 -61239.48569364 -46981.20596758], d=115321.767509

k=89: x=[32929.32366125 69339.2178524 53201.07825898], d=130578.703546

k=90: x=[-37283.70881022 -78515.21638603 -60235.24430461], d=147854.434238

k=91: x=[42218.52071074 88900.59093626 68208.83949208], d=167415.807322

k=92: x=[-47801.96254359 -100664.24335087 -77228.57337988], d=189564.834287

k=93: x=[54128.27585517 113980.27679449 87450.17314905], d=214644.520145

k=94: x=[-61287.38412952 -129061.93027164 -99015.77655966], d=243042.207066

k=95: x=[69397.89877778 146134.75995841 112119.79373611], d=275196.690230

k=96: x=[-78577.11264064 -165470.80137502 -126949.04466325], d=311605.561333

k=97: x=[88975.09209248 187360.40589932 143748.87919143], d=352831.207274

k=98: x=[-100744.43117161 -212150.48917847 -162762.57744761], d=399510.895078

k=99: x=[114075.08883747 240216.14723885 184300.47467676], d=452366.636417

k=100: x=[-129165.2148135 -271998.88067312 -208679.38263851], d=512215.027912

x = [-129165.2148135 -271998.88067312 -208679.38263851]

d = 512215.027912

count = 100

Zidel.py(上から、講義資料、(1)、(2))

A = [[5. -4.]

[-1. 2.]]

b = [4. 1.]

x0 = [0. 0.]

k=00: x=[0. 0.], d=-

k=01: x=[0.8 0.9], d=0.900000

k=02: x=[1.52 1.26], d=0.720000

k=03: x=[1.808 1.404], d=0.288000

k=04: x=[1.9232 1.4616], d=0.115200

k=05: x=[1.96928 1.48464], d=0.046080

k=06: x=[1.987712 1.493856], d=0.018432

k=07: x=[1.9950848 1.4975424], d=0.007373

k=08: x=[1.99803392 1.49901696], d=0.002949

k=09: x=[1.99921357 1.49960678], d=0.001180

k=10: x=[1.99968543 1.49984271], d=0.000472

k=11: x=[1.99987417 1.49993709], d=0.000189

k=12: x=[1.99994967 1.49997483], d=0.000075

x = [1.99994967 1.49997483]

d = 0.000075

count = 12

A = [[4.1 1.4]

[8.2 3.8]]

b = [11.77 28.14]

x0 = [0. 0.]

k=00: x=[0. 0.], d=-

k=01: x=[2.87073171 1.21052632], d=2.870732

k=02: x=[2.45738126 2.10249307], d=0.891967

k=03: x=[2.15280724 2.75973174], d=0.657239

k=04: x=[1.92838428 3.24401286], d=0.484281

k=05: x=[1.76302 3.60085158], d=0.356839

k=06: x=[1.64117263 3.86378538], d=0.262934

k=07: x=[1.55139036 4.05752607], d=0.193741

k=08: x=[1.485235 4.20028236], d=0.142756

k=09: x=[1.43648895 4.30547122], d=0.105189

k=10: x=[1.4005708 4.38297879], d=0.077508

k=11: x=[1.3741048 4.44008964], d=0.057111

k=12: x=[1.35460354 4.48217131], d=0.042082

k=13: x=[1.34023419 4.51317886], d=0.031008

k=14: x=[1.32964624 4.53602653], d=0.022848

k=15: x=[1.3218446 4.55286165], d=0.016835

k=16: x=[1.31609602 4.56526648], d=0.012405

k=17: x=[1.31186023 4.57440688], d=0.009140

k=18: x=[1.30873911 4.58114191], d=0.006735

k=19: x=[1.30643935 4.58610457], d=0.004963

k=20: x=[1.30474478 4.58976126], d=0.003657

k=21: x=[1.30349616 4.59245567], d=0.002694

k=22: x=[1.30257611 4.59444102], d=0.001985

k=23: x=[1.30189819 4.59590391], d=0.001463

k=24: x=[1.30139867 4.59698183], d=0.001078

k=25: x=[1.3010306 4.59777608], d=0.000794

k=26: x=[1.30075939 4.59836132], d=0.000585

k=27: x=[1.30055955 4.59879255], d=0.000431

k=28: x=[1.3004123 4.5991103], d=0.000318

k=29: x=[1.3003038 4.59934443], d=0.000234

k=30: x=[1.30022385 4.59951695], d=0.000173

k=31: x=[1.30016494 4.59964407], d=0.000127

k=32: x=[1.30012154 4.59973774], d=0.000094

x = [1.30012154 4.59973774]

d = 0.000094

count = 32

A = [[8. 2. 3.]

[4. 1. -1.]

[1. 3. 4.]]

b = [12. 1. 6.]

x0 = [0. 0. 0.]

k=00: x=[0. 0. 0.], d=-

k=01: x=[1.5 -5. 4.875], d=5.000000

k=02: x=[0.921875 2.1875 -0.37109375], d=7.187500

k=03: x=[1.09228516 -3.74023438 4.03210449], d=5.927734

k=04: x=[0.92301941 1.34002686 0.26422501], d=5.080261

k=05: x=[1.06590891 -2.99941063 3.48308074], d=4.339437

k=06: x=[0.94369738 0.70829123 0.73285723], d=3.707702

k=07: x=[1.04810573 -2.45956569 3.08264783], d=3.167857

k=08: x=[0.95889848 0.2470539 1.07498496], d=2.706620

k=09: x=[1.03511717 -2.06548371 2.79033349], d=2.312538

k=10: x=[0.96999587 -0.08964998 1.32473852], d=1.975834

k=11: x=[1.02563555 -1.77780368 2.57694388], d=1.688154

k=12: x=[0.97809697 -0.335444 1.50705875], d=1.442360

k=13: x=[1.01871397 -1.56779711 2.42116934], d=1.232353

k=14: x=[0.98401077 -0.51487376 1.64015263], d=1.052923

k=15: x=[1.01366121 -1.4144922 2.30745385], d=0.899618

k=16: x=[0.98832786 -0.64585758 1.73731122], d=0.768635

k=17: x=[1.00997269 -1.30257953 2.22444147], d=0.656722

k=18: x=[0.99147933 -0.74147584 1.80823705], d=0.561104

k=19: x=[1.00728007 -1.22088322 2.1638424], d=0.479407

k=20: x=[0.99377991 -0.81127723 1.86001294], d=0.409606

k=21: x=[1.00531445 -1.16124487 2.11960504], d=0.349968

k=22: x=[0.99545933 -0.86223227 1.89780937], d=0.299013

k=23: x=[1.00387955 -1.11770884 2.08731174], d=0.255477

k=24: x=[0.99668531 -0.89942949 1.92540079], d=0.218279

k=25: x=[1.00283208 -1.08592752 2.06373762], d=0.186498

k=26: x=[0.99758027 -0.92658347 1.94554253], d=0.159344

k=27: x=[1.00206742 -1.06272713 2.0465285], d=0.136144

k=28: x=[0.9982336 -0.94640589 1.96024602], d=0.116321

k=29: x=[1.00150922 -1.04579084 2.03396583], d=0.099385

k=30: x=[0.99871053 -0.96087627 1.97097957], d=0.084915

k=31: x=[1.00110173 -1.03342734 2.02479507], d=0.072551

k=32: x=[0.99905868 -0.97143966 1.97881507], d=0.061988

k=33: x=[1.00080426 -1.02440198 2.01810042], d=0.052962

k=34: x=[0.99931284 -0.97915094 1.98453499], d=0.045251

k=35: x=[1.00058711 -1.01781346 2.01321331], d=0.038663

k=36: x=[0.99949837 -0.98478017 1.98871054], d=0.033033

k=37: x=[1.00042859 -1.01300383 2.00964573], d=0.028224

k=38: x=[0.99963381 -0.98888952 1.99175869], d=0.024114

k=39: x=[1.00031287 -1.0094928 2.00704139], d=0.020603

k=40: x=[0.99973268 -0.99188934 1.99398384], d=0.017603

k=41: x=[1.0002284 -1.00692975 2.00514021], d=0.015040

k=42: x=[0.99980486 -0.99407921 1.9956082], d=0.012851

k=43: x=[1.00016673 -1.00505872 2.00375236], d=0.010980

k=44: x=[0.99985755 -0.99567782 1.99679398], d=0.009381

k=45: x=[1.00012171 -1.00369287 2.00273922], d=0.008015

k=46: x=[0.99989601 -0.99684481 1.9976596], d=0.006848

k=47: x=[1.00008885 -1.0026958 2.00199964], d=0.005851

k=48: x=[0.99992409 -0.99769671 1.99829151], d=0.004999

k=49: x=[1.00006486 -1.00196793 2.00145973], d=0.004271

k=50: x=[0.99994458 -0.9983186 1.9987528], d=0.003649

k=51: x=[1.00004735 -1.00143659 2.00106561], d=0.003118

k=52: x=[0.99995955 -0.99877257 1.99908954], d=0.002664

k=53: x=[1.00003456 -1.00104871 2.00077789], d=0.002276

k=54: x=[0.99997047 -0.99910398 1.99933537], d=0.001945

k=55: x=[1.00002523 -1.00076556 2.00056786], d=0.001662

k=56: x=[0.99997844 -0.9993459 1.99951482], d=0.001420

k=57: x=[1.00001842 -1.00055886 2.00041454], d=0.001213

k=58: x=[0.99998426 -0.99952251 1.99964582], d=0.001036

k=59: x=[1.00001345 -1.00040797 2.00030261], d=0.000885

k=60: x=[0.99998851 -0.99965143　1.99974145], d=0.000757

k=61: x=[1.00000982 -1.00029782　2.00022091], d=0.000646

k=62: x=[0.99999161 -0.99974554　1.99981126], d=0.000552

k=63: x=[1.00000717 -1.00021741　2.00016126], d=0.000472

k=64: x=[0.99999388 -0.99981425　1.99986222], d=0.000403

k=65: x=[1.00000523 -1.00015871　2.00011772], d=0.000344

k=66: x=[0.99999553 -0.9998644　 1.99989942], d=0.000294

k=67: x=[1.00000382 -1.00011586　2.00008594], d=0.000251

k=68: x=[0.99999674 -0.99990101　1.99992657], d=0.000215

k=69: x=[1.00000279 -1.00008458　2.00006273], d=0.000184

k=70: x=[0.99999762 -0.99992774　1.9999464], d=0.000157

k=71: x=[1.00000203 -1.00006174　2.0000458], d=0.000134

k=72: x=[0.99999826 -0.99994725　1.99996087], d=0.000114

k=73: x=[1.00000149 -1.00004507　2.00003343], d=0.000098

x =　[1.00000149 -1.00004507　2.00003343]

d = 0.000098

count = 73

分かりにくいので、収束までの回数をまとめると、このようになる。

問題(テスト例)	講義資料	(1)	(2)
ヤコビ法	21 回	75 回	100 回以上
ガウスザイデル法	12 回	32 回	73 回

明らかにガウスサイデル法の方が早く収束すると考察できる。

(3)ガウスザイデル法の収束条件について。

一言で言うなら係数行列が狭義対角優勢行列である必要がある。例を挙げると、

収束するケースは｛$3x + y = 6$, $x + 2y = 7$、しないケースは｛$x + 3y = 10$, $2x + y = 5$ 。

実際に行列に変換して代入すると、

Zidel.py(収束ケース、非収束ケースの順)

A = [[3. 1.]

[1. 2.]]

b = [6. 7.]

x0 = [0. 0.]

k=00: x=[0. 0.], d=-

k=01: x=[2. 2.5], d=2.500000

k=02: x=[1.16666667 2.91666667], d=0.833333

k=03: x=[1.02777778 2.98611111], d=0.138889

k=04: x=[1.00462963 2.99768519], d=0.023148

k=05: x=[1.0007716 2.9996142], d=0.003858

k=06: x=[1.0001286 2.9999357], d=0.000643

k=07: x=[1.00002143 2.99998928], d=0.000107

k=08: x=[1.00000357 2.99999821], d=0.000018

x = [1.00000357 2.99999821]

d = 0.000018

count = 8

A = [[1. 3.]

[2. 1.]]

b = [10. 5.]

```
x0 =    [0. 0.]


k=00: x=[0. 0.], d=-

k=01: x=[ 10. -15.], d=15.000000

k=02: x=[    55. -105.], d=90.000000

k=03: x=[ 325. -645.], d=540.000000

k=04: x=[ 1945. -3885.], d=3240.000000

k=05: x=[ 11665. -23325.], d=19440.000000

k=06: x=[    69985. -139965.], d=116640.000000

k=07: x=[ 419905. -839805.], d=699840.000000

k=08: x=[ 2519425. -5038845.], d=4199040.000000

k=09: x=[ 15116545. -30233085.], d=25194240.000000

k=10: x=[ 9.06992650e+07 -1.81398525e+08], d=151165440.000000

k=11: x=[ 5.44195585e+08 -1.08839116e+09], d=906992640.000000

k=12: x=[ 3.2651735e+09 -6.5303470e+09], d=5441955840.000000

k=13: x=[ 1.9591041e+10 -3.9182082e+10], d=32651735040.000000

k=14: x=[ 1.17546246e+11 -2.35092492e+11], d=195910410240.000000

k=15: x=[ 7.05277477e+11 -1.41055495e+12], d=1175462461440.000000

k=16: x=[ 4.23166486e+12 -8.46332972e+12], d=7052774768640.000000

k=17: x=[ 2.53899892e+13 -5.07799783e+13], d=42316648611840.000000

k=18: x=[ 1.52339935e+14 -3.04679870e+14], d=253899891671040.000000

k=19: x=[ 9.14039610e+14 -1.82807922e+15], d=1523399350026240.000000

k=20: x=[ 5.48423766e+15 -1.09684753e+16], d=9140396100157440.000000

k=21: x=[ 3.29054260e+16 -6.58108519e+16], d=54842376600944640.000000
```

k=22: x=[1.97432556e+17 -3.94865112e+17], d=329054259605667776.000000

k=23: x=[1.18459533e+18 -2.36919067e+18], d=1974325557634006528.000000

k=24: x=[7.10757201e+18 -1.42151440e+19], d=11845953345804038144.000000

k=25: x=[4.26454320e+19 -8.52908641e+19], d=71075720074824220672.000000

k=26: x=[2.55872592e+20 -5.11745185e+20], d=426454320448945389568.000000

k=27: x=[1.53523555e+21 -3.07047111e+21], d=2558725922693672075264.000000

k=28: x=[9.21141332e+21 -1.84228266e+22], d=15352355536162034548736.000000

k=29: x=[5.52684799e+22 -1.10536960e+23], d=92114133216972198903808.000000

k=30: x=[3.31610880e+23 -6.63221759e+23], d=552684799301833126313984.000000

k=31: x=[1.98966528e+24 -3.97933055e+24], d=3316108795810999294754816.000000

k=32: x=[1.19379917e+25 -2.38759833e+25], d=19896652774865995768528896.000000

k=33: x=[7.162795e+25 -1.432559e+26], d=119379916649195983201107968.000000

k=34: x=[4.297677e+26 -8.595354e+26], d=716279499895175967926124544.000000

k=35: x=[2.5786062e+27 -5.1572124e+27], d=4297676999371056357312561152.000000

k=36: x=[1.54716372e+28 -3.09432744e+28], d=25786061996226338143875366912.000000

k=37: x=[9.28298232e+28 -1.85659646e+29], d=154716371977358028863252201472.000000

k=38: x=[5.56978939e+29 -1.11395788e+30], d=928298231864148243548257386496.000000

k=39: x=[3.34187363e+30 -6.68374727e+30],
d=5569789391184889742764521029632.000000

k=40: x=[2.00512418e+31 -4.01024836e+31],
d=33418736347109338456587126177792.000000

k=41: x=[1.20307451e+32 -2.40614902e+32],
d=200512418082656039746722011807744.000000

k=42: x=[7.21844705e+32 -1.44368941e+33],
d=1203074508495936238480332070846464.000000

k=43: x=[4.33106823e+33 -8.66213646e+33],
d=7218447050975617719112368576790528.000000

k=44: x=[2.59864094e+34 -5.19728188e+34],
d=433106823058537017029881930333552 64.000000

k=45: x=[1.55918456e+35 -3.11836913e+35],
d=2598640938351222102179291582001315 84.000000

k=46: x=[9.35510738e+35 -1.87102148e+36],
d=15591845630107332613075749492007895 04.000000

k=47: x=[5.61306443e+36 -1.12261289e+37],
d=93551073780643989775496393364990853 12.000000

k=48: x=[3.36783866e+37 -6.73567731e+37],
d=561306442683863891429313531493492981 76.000000

k=49: x=[2.02070319e+38 -4.04140639e+38],
d=3367838656103183537470540503746766438 40.000000

k=50: x=[1.21242192e+39 -2.42484383e+39],
d=20207031936619099713665968504194130247 68.000000

k=51: x=[7.2745315e+39 -1.4549063e+40],
d=121242191619714598281995811025164781486 08.000000

k=52: x=[4.3647189e+40 -8.7294378e+40],
d=727453149718287686406040435321322665410 56.000000

k=53: x=[2.61883134e+41 -5.23766268e+41],
d=4364718898309725344723718058565264180510 72.000000

k=54: x=[1.57129880e+42 -3.14259761e+42],
d=26188313389858353615767357458116928706969 60.000000

k=55: x=[9.42779282e+42 -1.88555856e+43],
d=157129880339150121694604144748701572241817 60.000000

k=56: x=[5.65667569e+43 -1.13133514e+44],
d=942779282034900878720429582737842421345812 48.000000

k=57: x=[3.39400542e+44 -6.78801083e+44],
d=5656675692209405272322577496427054528074874 88.000000

k=58: x=[2.03640325e+45 -4.07280650e+45],
d=33940054153256431633935464978562327168449249 28.000000

k=59: x=[1.22184195e+46 -2.44368390e+46],
d=203640324919538564450600785306785933076631388 16.000000

k=60: x=[7.33105170e+46 -1.46621034e+47],
d=1221841949517231589527700748357419837932301189 12.000000

k=61: x=[4.39863102e+47 -8.79726204e+47],
d=7331051697103389131518012417111110548648781414 40.000000

k=62: x=[2.63917861e+48 -5.27835722e+48],
d=43986310182620334789108074502666663291892688486 40.000000

k=63: x=[1.58350717e+49 -3.16701433e+49],
d=263917861095722047676874886027207193730078600069 12.000000

k=64: x=[9.5010430e+49 -1.9002086e+50],
d=1583507166574332493753123657556348303600324769218 56.000000

k=65: x=[5.70062580e+50 -1.14012516e+51],
d=9501042999445994131751244579765669256722535940096 00.000000

k=66: x=[3.42037548e+51 -6.84075096e+51],
d=57006257996675954821297499091724968761782263537991 68.000000

k=67: x=[2.05222529e+52 -4.10445058e+52],
d=342037547980055728927784994550349812570693581227950 08.000000

k=68: x=[1.23133517e+53 -2.46267035e+53],
d=2052225287880334586243189292888638540033291132222832 64.000000

k=69: x=[7.38801104e+53 -1.47760221e+54],
d=12313351727282007517459135757331831240199746793336995 84.000000

k=70: x=[4.43280662e+54 -8.86561324e+54],
d=73880110363692045104754814543990987441198480760021975 04.000000

k=71: x=[2.65968397e+55 -5.31936795e+55],
d=4432806621821522161833501799137917705072536954772180172 8.000000

k=72: x=[1.59581038e+56 -3.19162077e+56],
d=2659683973092913732661530738283983856163019685526618767 36.000000

k=73: x=[9.57486230e+56 -1.91497246e+57],
d=1595810383855748152484632511210143667073912308783309127680.000000

k=74: x=[5.74491738e+57 -1.14898348e+58],
d=9574862303134489611806082521342835175434669872961151827968.000000

k=75: x=[3.44695043e+58 -6.89390086e+58],
d=5744917381880693209565019549540122566867845107567653447 2704.000000

k=76: x=[2.06817026e+59 -4.13634051e+59],
d=3446950429128416148746463715030304955477889791024207128 16640.000000

k=77: x=[1.24090215e+60 -2.48180431e+60],
d=2068170257477049510841916640773197841000987693427632229056512.000000

k=78: x=[7.44541293e+60 -1.48908259e+61],
d=1240902154486229706505149984463918704600592616056579337 4339072.000000

k=79: x=[4.46724776e+61 -8.93449551e+61],
d=7445412926917378524480438447975488439260749586238503301 1527680.000000

k=80: x=[2.68034865e+62 -5.36069731e+62],
d=4467247756150427799767155567646035971533715087500767444 41004032.000000

k=81: x=[1.60820919e+63 -3.21641838e+63],
d=2680348653690256497172588674224756807459624962965083009654456320.000000

k=82: x=[9.64925515e+63 -1.92985103e+64],
d=1608209192214153898303553204534854084475774977779049805792673 7920.000000

k=83: x=[5.78955309e+64 -1.15791062e+65],
d=9649255153284923389821319227209124506854649866674298834756042 7520.000000

k=84: x=[3.47373186e+65 -6.94746371e+65],
d=5789553091970953332372005617492073966344070216188729866006005 02272.000000

k=85: x=[2.08423911e+66 -4.16847823e+66],
d=34737318551825727477120416839175385000930764804501439834410650173 44.000000

k=86: x=[1.25054347e+67 -2.50108694e+67],
d=20842391131095434989694573476660642759985190181227051772971466096640.00000
0

k=87: x=[7.50326081e+67 -1.50065216e+68],
d=12505434678657259197923532133782879767303191666967656510572970849075 2.0000
00

k=88: x=[4.50195648e+68 -9.00391297e+68],
d=7503260807194354560944406239089191386415023031237354144631831144693 76.0000
00

k=89: x=[2.70117389e+69 -5.40234778e+69],
d=4501956484316613119690528959925729421435770606319708391463879232716800.000
000

k=90: x=[1.62070433e+70 -3.24140867e+70],
d=2701173890589967412065655116188780145357354218699069949256590884549427 2.00
0000

k=91: x=[9.72422601e+70 -1.94484520e+71],
d=1620704334353980692438679608255485424549936875268911348552214080106004 48.0
00000

k=92: x=[5.83453560e+71 -1.16690712e+72],
d=9724226006123883173834931495364043197957523875415590575320246283130634 24.0
00000

k=93: x=[3.50072136e+72 -7.00144272e+72],
d=5834535603674330296619817358885973658511353275728505351589363048880537 600.
000000

k=94: x=[2.10043282e+73 -4.20086563e+73],
d=3500721362204598177971890415331584195106811965437103210953617829328322 5600
.000000

k=95: x=[1.26025969e+74 -2.52051938e+74],
d=2100432817322758530157030125998104686916721786802276960430844029754922762 4.000000

k=96: x=[7.56155814e+74 -1.51231163e+75],
d=1260259690393655218527845841785755033522663843404028833896193529277505863680.000000

k=97: x=[4.53693489e+75 -9.07386977e+75],
d=7561558142361932516370608244957236857607552316296124895029406512759661658112.000000

k=98: x=[2.72216093e+76 -5.44432186e+76],
d=45369348854171588670471472433782318977796944533126339281364463945386628743168.000000

k=99: x=[1.63329656e+77 -3.26659312e+77],
d=272216093125029557733837542746538322538175144657359676043434684197005137281024.000000

k=100: x=[9.79977935e+77 -1.95995587e+78],
d=1633296558750177552091094921629985204600198687612971179102591309379513742262272.000000

x = [9.79977935e+77 -1.95995587e+78]

d =
1633296558750177552091094921629985204600198687612971179102591309379513742262272.000000

count = 100

となり、動作として確認できる。

2.3 課題 3

適当な連立一次方程式の問題を作って解く。

私はこのような問題を考えた。「今ここに、金、銀、銅の 3 種類のコインがある。天秤を使って、組み合わせを試した所、金 6 枚と銀 2 枚は銅 2 枚と釣り合い、金 3 枚と銅 4 枚は銀 7

枚と釣り合った。ここで、銅のコインが 80g だと仮定すると、他 2 種のコインの重さはどうなるか」。この問題をプログラムで解くと、答えは金が 10g、銀が 50g となる。なお、プログラムへの入力は、A = [[6,2],[3,-7]]、b = [160, -320]とした。

以下に実行結果を示すが、近似値系は結果の部分だけを示す。

```
# Gauss_nopivot.py
初期値(1)
[[     6.       2.    160.]
 [     3.     -7. -320.]]

k =    0 , pivot =    6.0
[[     6.       2.    160.]
 [     0.     -8. -400.]]

解 x =   [10. 50.]
# Gauss.py
初期値(2)
[[     6.       2.    160.]
 [     3.     -7. -320.]]

k =    0 , pivot =    6.0
[[     6.       2.    160.]
 [     0.     -8. -400.]]

解 x =   [10. 50.]
# Jacobi.py
x =   [ 9.99997976 50.0000052 ]
```

d = 0.000056

count = 15

Zidel.py

x = [10.00000289 50.00000124]

d = 0.000023

count = 9

のように、正しく求められた。

3.発展課題

3.1 発展課題 1

係数行列の性質についてまとめた上で、実際に値を求め、理由を説明する。

行列ノルムは、行列の「大きさ」を示す尺度であり。L2 ノルムは、特にスペクトルノルムとも呼ばれ、行列の最大特異値に等しくなる。行列の条件数は、連立一次方程式を解く際の解の安定性や誤差の増幅のしやすさを示す指標であり、条件数が大きいほど、わずかな入力の誤差が解に大きな誤差となって現れやすくなる。

本題に入る。プログラム名は"hatten1.py"であり、例として課題 1 の(2)の行列について値を求めると、L2 ノルムが 9.948935187737995、条件数が 8.154283294325825 となる。L2 ノルムの方は、行列の最大特異値だから、としか言えない。条件数の方は、1 に近ければ近いほど安定した良い条件であり(最悪 10000 程度の値を取る可能性があることも考慮すると)、誤差はさほど大きくなく、悪くない行列だから、と言える。

総じて、その行列の安定性を示すのが「ノルム」と「条件数」である。

3.2 発展課題 2

未実装。

4.感想と考察

感想としては、シンプルに難化したと同時に、実践的になったと感じた。

今回の考察としては、連立一次方程式の解法という基礎的ながら重要な課題に対して、行列表現や LU 分解といった理論的な背景、ピボッティングや収束性といった実装における問題点、そしてアルゴリズムも含めた解法などの様々な観点からアプローチした回であったと思われる。